



## Full Length Article

## Cellular Therapy

## Human Amniotic Membrane-Derived Mesenchymal Stem Cells Prevent Acute Graft-Versus-Host Disease in an Intestinal Microbiome-Dependent Manner

Xiaoyin Bu<sup>1,2,†</sup>, Ya Gao<sup>3,†</sup>, Weifeng Pan<sup>1</sup>, Liping Liu<sup>1</sup>, Junhui Wang<sup>1</sup>, Zhao Yin<sup>1</sup>, Baohong Ping<sup>2,1,\*</sup>

<sup>1</sup> Department of Hematology, Nanfang Hospital, Southern Medical University, Guangzhou, China

<sup>2</sup> Department of Hematology, Huiqiao Medical Center, Nanfang Hospital, Southern Medical University, Guangzhou, China

<sup>3</sup> Department of Laboratory Medicine, Nanfang Hospital, Southern Medical University, Guangzhou, 510515, China

## Article history:

Received 13 August 2023

Accepted 3 November 2023

## Key Words:

Acute graft-versus-host disease

Human amniotic membrane-derived mesenchymal stem cells

Intestinal microbiome

Intestinal homeostasis

Short-chain fatty acids

## A B S T R A C T

Acute graft-versus-host disease (aGVHD) represents a fatal severe complication after allogeneic hematopoietic stem cell transplantation. As a promising cell therapeutic strategy of aGVHD, the mechanism of mesenchymal stem cells (MSC) to ameliorate aGVHD has not been fully clarified, especially in the field of intestinal homeostasis including the intestinal microbiome involved in the pathogenesis of aGVHD. The present study aimed to explore the effect of MSC on intestinal homeostasis including the intestinal barrier and intestinal microbiome and its metabolites, as well as the role of intestinal microbiome in the preventive process of hAMSCs ameliorating aGVHD. The preventive effects of human amniotic membrane-derived MSC (hAMSCs) was assessed in humanized aGVHD mouse models. Immunohistochemistry and RT-qPCR were used to evaluate intestinal barrier function. The 16S rRNA sequencing and targeted metabolomics assay were performed to observe the alternation of intestinal microbiome and the amounts of medium-chain fatty acids (MCFAs) and short-chain fatty acids (SCFAs), respectively. Flow cytometry was performed to analyze the frequencies of T immune cells. Through animal experiments, we found that hAMSCs had the potential to prevent aGVHD. hAMSCs could repair the damage of intestinal barrier structure and function, as well as improve the dysbiosis of intestinal microbiome induced by aGVHD, and meanwhile, upregulate the concentration of metabolites SCFAs, so as to reshape intestinal homeostasis. Gut microbiota depletion and fecal microbial transplantation confirmed the involvement of intestinal microbiome in the preventive process of hAMSCs on aGVHD. Our findings showed that hAMSCs prevented aGVHD in an intestinal microbiome-dependent manner, which might shed light on a new

*Financial disclosure:* See Acknowledgments on page 189.e12.

\*Correspondence and reprint requests: Baohong Ping, MMed, Department of Hematology, Huiqiao Medical Center, Nanfang Hospital, Southern Medical University, Guangzhou, 510515, China.

E-mail address: [nfyypingbaohong66@163.com](mailto:nfyypingbaohong66@163.com) (B. Ping).

† Xiaoyin Bu and Ya Gao contributed equally to this work.

<https://doi.org/10.1016/j.jtct.2023.11.005>

2666-6367/© 2023 The American Society for Transplantation and Cellular Therapy. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

mechanism of hAMSCs inhibiting aGVHD and promote the development of new prophylaxis regimes for aGVHD prevention.

© 2023 The American Society for Transplantation and Cellular Therapy. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Acute graft-versus-host disease (aGVHD), which is due to donor T lymphocytes recognizing the host as foreign and thus initiating the pathogenic immune response in the recipient and causing a cytokine storm, remains the crucial contributing factor for nonrecurrent death of allogeneic hematopoietic stem cell transplantation (allo-HSCT) [1,2]. With the development of 16S rRNA sequencing, metagenomics, and targeted metabolomics, the role of the intestinal microbiome in the disease pathogenesis has attracted a great deal of attention [3,4], and intestinal damage was not just the result of immune attack but rather a result of multiple factors [5]. Conditioning regimen could damage epithelial cells and lead to the loss of intestinal barrier [6]. On the other hand, a conditioning regimen and the application of antibiotics can lead to intestinal microbiome dysbiosis [6]. Impaired intestinal homeostasis including damaged intestinal barrier and altered intestinal microbiome contributes to the translocation of gut commensals and pathogen-associated-molecular patterns, which were recognized by antigen-presenting cells, and further activate T lymphocytes and trigger cytokine storm to accelerate aGVHD [5,7]. Moreover, bacteria itself and its metabolites such as short-chain fatty acids (SCFAs), indoles, and bile acids (BAs), play vital roles in maintaining intestinal homeostasis and shaping intestinal mucosal immunity, whereas in the context of aGVHD, they are disturbed to damage both in the intestinal barrier and immune cells [5]. Consequently, remodeling intestinal homeostasis, especially the intestinal microbiome, may be a potential new option for aGVHD treatment.

The capabilities of mesenchymal stem cells (MSC) in immunosuppression and tissue repair enable them as an eligible and ideal cellular therapeutic candidate for the treatment of various immune disorders, including aGVHD [8]. Human amniotic membrane-derived MSC (hAMSCs), because of their availability, abundant cell resources, high proliferation ability, low immunogenicity, and no tumorigenicity, have become a promising new source of MSC [9,10]. Clinical trials and preclinical studies have shown that hAMSCs have been widely applied in

the treatment of psoriasiform dermatitis, arthritis, colitis, and sclerosing cholangitis [9]. Previous studies including ours have shown that hAMSCs have the ability to attenuate aGVHD [11–13], and furthermore, we found that the disruption of intestinal microbiome resulting from aGVHD could be improved after hAMSCs infusion [14]. However, the study on the signature of intestinal microbiome and its metabolites after the prevention of aGVHD with hAMSCs is still insufficient, and the role of intestinal microbiome in the preventive process of hAMSCs alleviating aGVHD has not been fully elucidated. Consequently, in the present study, we aim to explore the impact of hAMSCs on the intestinal microbiome and its metabolites, as well as the role of intestinal microbiome in the underlying mechanism.

## MATERIAL AND METHODS

### *Isolation and Collection of Human Peripheral Blood Mononuclear Cells*

Peripheral blood extracted from healthy donors with written informed consent was procured to isolate human peripheral blood mononuclear cells (PBMC) with Ficoll-Hypaque (Tianjin Haoyang, China) density centrifugation as previously shown [11].

### *Acute GVHD Inducing and hAMSCs Administration*

Eight- to 10-week-old NPG mice (weighing 25 to 30 g) were purchased from Beijing Vitalstar Biotechnology Co., Ltd., and raised in a specific pathogen-free environment under standard laboratory conditions. Thirty-five NPG mice were randomly divided into 3 groups, namely Control group (n = 5), aGVHD group (n = 15), and hAMSCs group (n = 15). To induce aGVHD, mice in the aGVHD and hAMSCs groups were given irradiation and then injected with PBMC as previously described [11]. hAMSCs  $5 \times 10^5$  were infused into aGVHD mice by tail vein injection in the hAMSCs group mice on the third day, whereas mice in the aGVHD group underwent tail vein injection of PBS. Five mice in the aGVHD group experienced severe aGVHD ( $>20\%$  weight loss or aGVHD clinical score  $>2$ ) and 5 mice in the hAMSCs group on day 6 after transplantation were euthanized to collect target organs, peripheral blood, and colon contents for further analysis; meanwhile, feces were collected before euthanization. Intestines, colon contents, and feces were also harvested from the mice in the control group. Additionally, to observe the aGVHD symptoms and survival of mice, 10 mice in the aGVHD group and 10 mice in the hAMSCs group were recorded daily for weight loss and other aGVHD symptoms, as well as survival until termination on day 28 or death. The aGVHD clinical score on the basis of the aGVHD clinical scoring system adapted from Cooke's initial description [15] was used to score the

aGVHD symptoms of mice. The experiment was repeated 3 times.

### **Gut Microbiota Depletion and Fecal Microbial Transplantation Experiments**

For gut microbiota depletion, 22 NPG mice aged 8 to 10 weeks ( $30 \pm 2$  g) were randomly divided into 2 groups, including the ABX(aGVHD) group ( $n = 11$ ) and the ABX(hAMSCs) group ( $n = 11$ ). Those mice were given quadruple antibiotic cocktails (vancomycin, 100 mg/kg; neomycin sulfate, 200 mg/kg; metronidazole, 200 mg/kg; ampicillin, 200 mg/kg) by oral gavage daily for 5 days (ABX mice). After aGVHD inducing, mice in ABX(aGVHD) group and ABX(hAMSCs) group were injected with PBS or  $5 \times 10^5$  hAMSCs, respectively. For fecal bacterial solution preparation, feces from each group of mice (aGVHD and hAMSCs groups) were weighed and homogenized in PBS (100 mg/mL). After filtering twice, the suspension was centrifuged at 3000g, at 4°C for 10 minutes. The precipitates were suspended with 2 mL PBS and centrifuged at 3000g at 4°C for 10 minutes again, then resuspended in PBS at 0.1 g/mL to prepare fecal bacterial solution. For fecal microbial transplantation (FMT), 28 antibiotic-treated mice were randomly divided into 2 groups, including the FMT(aGVHD) group ( $n = 14$ ) and FMT(hAMSCs) group ( $n = 14$ ) and then administered 200  $\mu$ L fecal bacterial solution prepared from the feces of aGVHD mice or hAMSCs-treated mice intragastrically daily, respectively, for 5 consecutive days. After FMT, those mice were procured to induce aGVHD. The aGVHD target organs and peripheral blood from all groups including ABX(aGVHD), ABX(hAMSCs), FMT(aGVHD), and FMT(hAMSCs) groups ( $n = 4$  per group) were collected for further analysis on the day 6 after aGVHD inducing. The remaining mice in all groups were used to investigate the body weight loss, aGVHD clinical score and survival. The evaluation of model construction of ABX mice and FMT mice is shown in [Supplementary Figure S1](#). The experiment was repeated 3 times.

### **Histopathological Assessment**

The aGVHD target organ tissues obtained at the time of euthanasia were fixed in 4% paraformaldehyde fixative, embedded in paraffin, sliced into 5  $\mu$ m sections, and then stained with hematoxylin and eosin (H&E). Histological scores of each target organs were scored using a previously described method [16]. For immunohistochemistry (IHC), the sections were incubated with primary antibodies (Abcam, Cambridge, MA) at 37°C for 1 hour. The bound antibodies were incubated with the secondary antibody goat anti-rabbit IgG antibody-HRP (Genentech Inc., San Francisco, CA) followed by the addition of DAB. Image J software was used for quantitative analysis.

### **The 16S rRNA Sequencing**

The DNA of feces was extracted according to the instructions of E.Z.N.A. stool DNA Kit (Omega Bio-Tek, Norcross, GA). DNA purity and concentration were determined using 1% agarose gel electrophoresis and NanoDrop One Spectrophotometer (Thermo Scientific, Swedesboro, NJ). The V3-V4 hypervariable regions was amplified with the bacterial 16S rDNA gene universal primers 338F (5'-ACTCCTACGGGAGGCAGCA-3'), and 806R (5'-GGACTACHVGGGTWTCTAAT-3') with barcodes and PremixTaq (TaKaRa Bio. Co., Shiga, Japan). Purified and pooled amplicons were sequenced on the Illumina Miseq platform according to the standard protocols by Magigene

Technology Co., Ltd. (Guangzhou, China) to obtain raw data in FASTQ format.

### **Targeted Metabolomics**

Colon contents from each group of mice (Control group, aGVHD group and hAMSCs group) were supplemented with 0.8 mL of extracting solution (25 mg/L stock in methyl tert-butyl ether) as internal standard, and then extracted according to the manufacturer's protocols (Magigene Technology Co., Ltd., Guangzhou, China). The extracted samples were detected using a GC2030-QP2020 NX gas chromatography-mass spectrometer (GC-MS) system (Shimadzu, Kyoto, Japan) equipped with a HP-FFAP capillary column. Medium-chain-fatty acids (MCFAs) and short-chain-fatty acids (SCFAs) standards were mixtures of acetate, propionate, butyrate, isobutyrate, valerate, isovalerate, caproate, heptate, octanoate, pelargonate and decanoate (Dr. Ehrenstorfer, Germany).

### **Quantitative Reverse Transcription Polymerase Chain Reaction (RT-qPCR)**

Trizol Reagent (TaKaRa, China) was used to extract total RNA from colonic tissue. cDNA synthesis was performed using total RNA as a template. RT-qPCR was performed on a LightCycler 480 PCR system (Roche, Switzerland) and the specific primer pairs were used as follows: ZO-1-F, 5'-ACCACCAACCCGAGAAGAC-3', and ZO-1-R, 5'-CAGGAGT-CATGGACGCACA-3'; Occludin-F, 5'-TTGAAAGTCCACCTCT-TACAGA-3', and Occludin-R, 5'-CCGGATAAAAAGAGTACGCTGG-3'; GAPDH-F, 5'-AGGTCGGTGTGAACGGATTG-3', and GAPDH-R, 5'-GGGGTCGTTGATGGCAACA-3'.

### **Flow Cytometry**

Single cell suspension was made from aGVHD target organs and peripheral blood. The cells were stained for surface markers and intracellular protein using antibodies against CD3, CD4, CD8, CD25, and Foxp3 (anti-CD3-PECy7, anti-CD4-FITC, anti-CD8-PE, anti-CD25-APC, anti-Foxp3-PE; BioLegend, San Diego, CA) according to the manufacturers' instructions. The stained cells were analyzed using BD FACS II Canto (BD Bioscience, Franklin Lakes, NJ) and FlowJo\_V10.

### **Statistical Analysis**

The data were analyzed and presented as the mean  $\pm$  SD with GraphPad Prism8.0. Independent Student's *t*-test and one-way analysis of variance (ANOVA) were used to compare between two groups or among multiple groups, respectively. The 16S rDNA sequencing data were analyzed on Magichand platform (<http://cloud.magigene.com/>) and Majorbio Cloud Platform (<https://cloud.majorbio.com/page/tools/>). Correlation matrix was plotted using R packages (v3.6.3). Statistical significance was indicated as follows: \*  $P < .05$ , \*\*  $P < .01$ , \*\*\*  $P < .001$ , \*\*\*\*  $P < .0001$ , and ns represented no significance.

## **RESULTS**

### **hAMSCs Prevented aGVHD in Humanized Mouse Models**

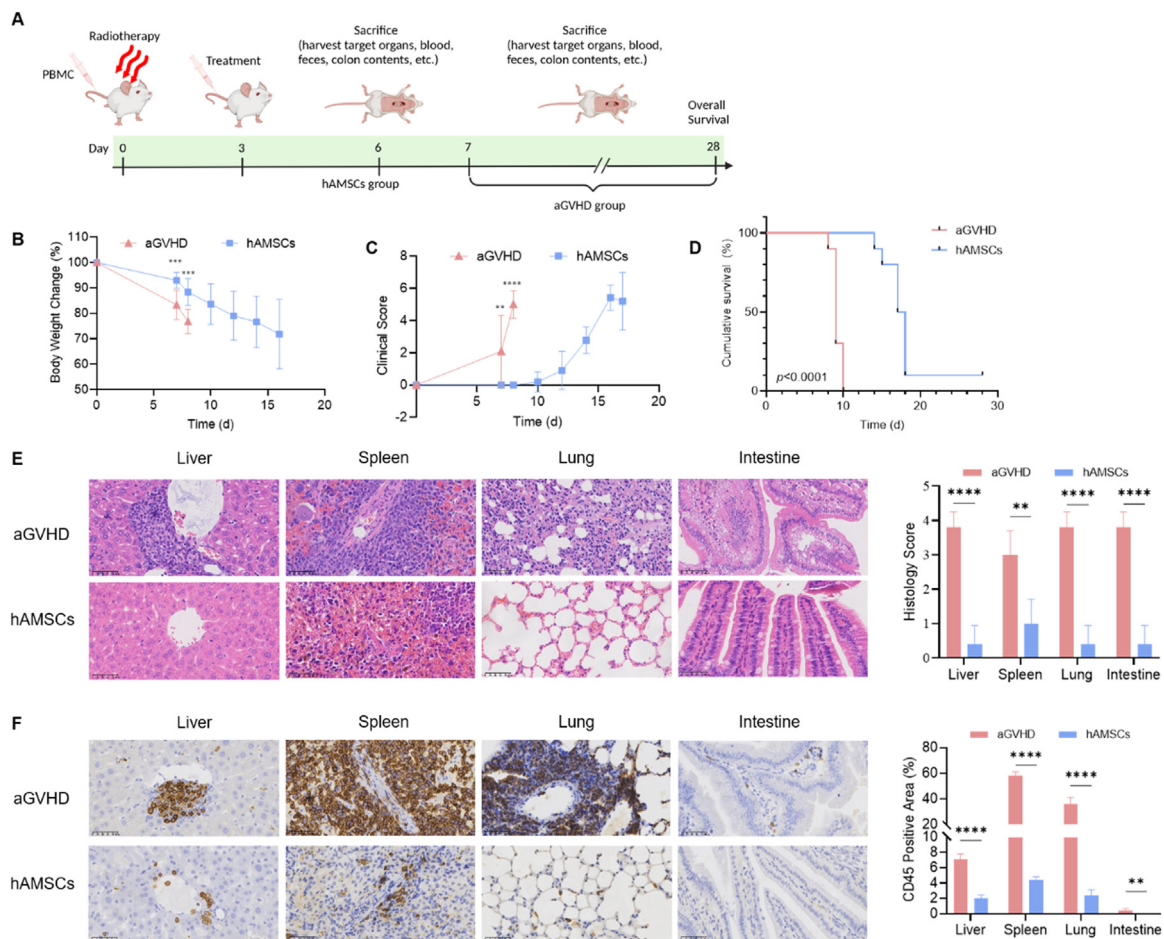
Human amniotic membrane-derived mesenchymal stem cells were isolated, cultured and identified as previously shown [14]. First, we investigated whether hAMSCs had a potential to



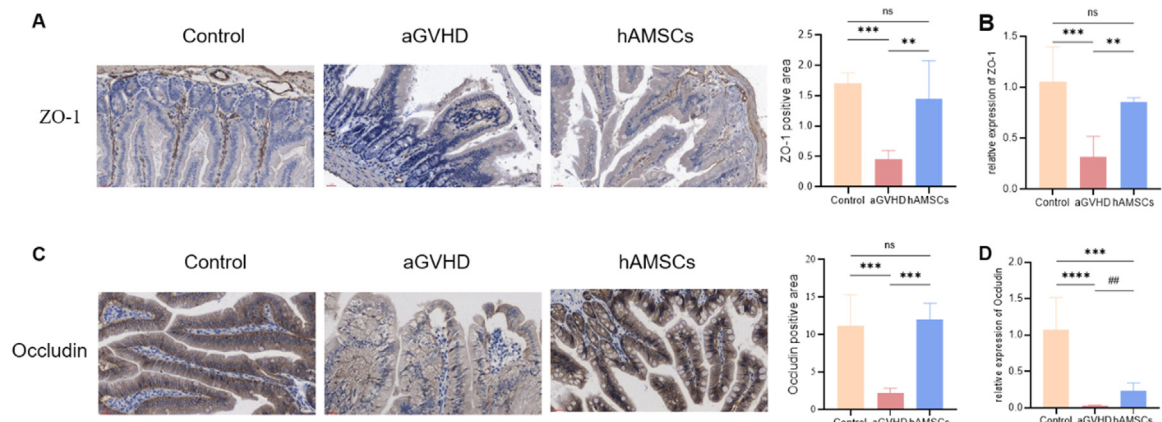
prevent aGVHD. Figure 1A represented a schematic diagram of aGVHD induction and hAMSCs infusion. As shown in Figure 1B, aGVHD group had a dramatically steeper weight loss curve, whereas the decline in the hAMSCs group was slower. Compared with the aGVHD group, hAMSCs-treated mice showed a later onset of aGVHD clinical score and a decrease in clinical score (Figure 1C). Additionally, hAMSCs administration resulted in prolonged survival, relative to the aGVHD group (Figure 1D). H&E staining of the aGVHD target organs revealed that aGVHD led to severe tissue injury and mass lymphocyte infiltration; however, all of these aGVHD pathological changes were improved by hAMSCs treatment (Figure 1E). Furthermore, results from IHC showed that hAMSCs treatment could reduce human CD45<sup>+</sup> cell infiltration in the aGVHD target organs (Figure 1F). Consequently, hAMSCs infusion could significantly inhibit aGVHD symptoms and prolong survival in humanized mouse models.

### ***HAMSCs Restored Impaired Intestinal Homeostasis Caused by aGVHD***

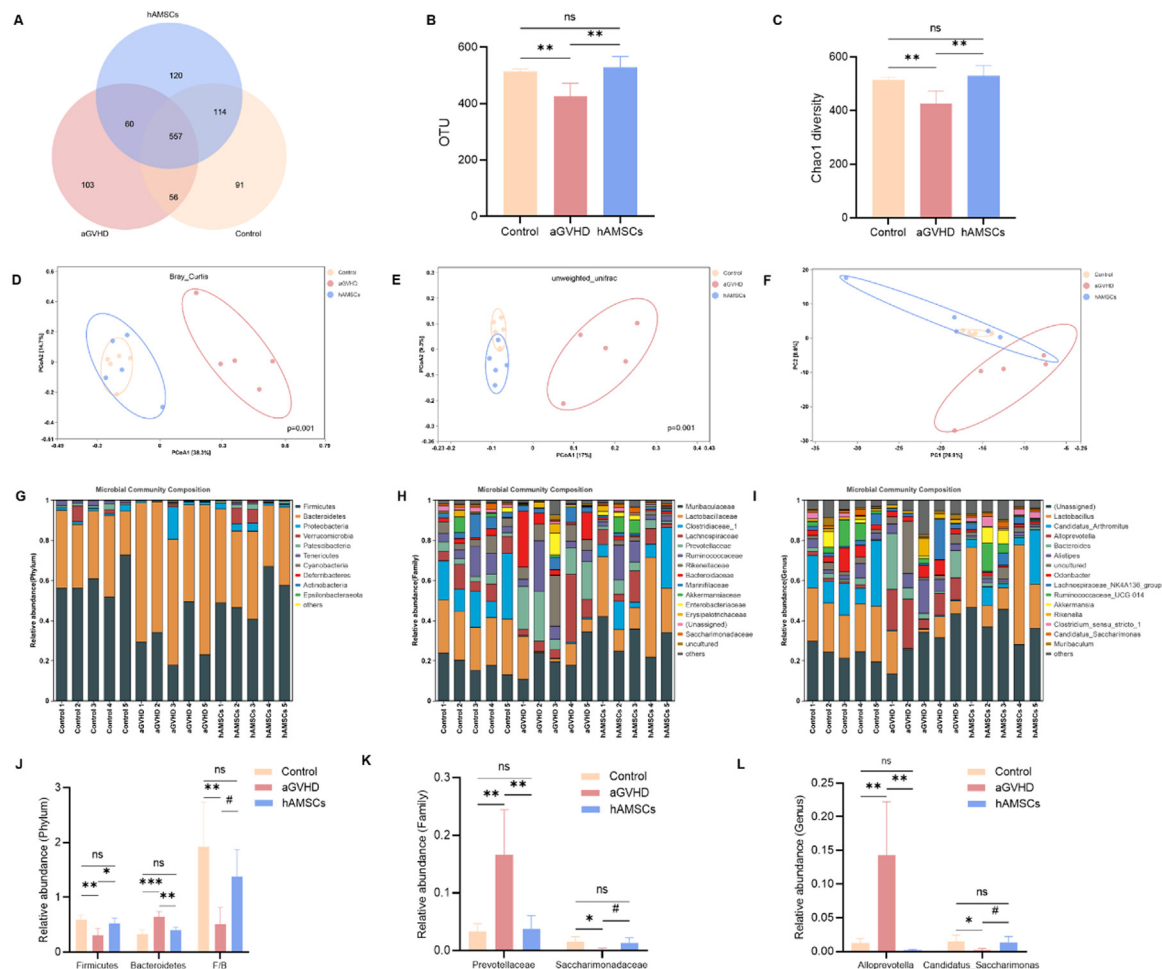
We observed that hAMSCs administration had the ability to improve the intestinal barrier structure destruction and exogenous T cells infiltration induced by aGVHD (Figure 1E,F); we then measured the intestinal tight junction proteins (TJs) to evaluate intestinal barrier integrity. IHC and RT-qPCR results showed that the protein and mRNA expression levels of TJs ZO-1 and Occludin in aGVHD group were dramatically lower than those in Control group, and the levels of these indicators were significantly reversed after hAMSCs treatment (Figure 2). We next explored the impact of hAMSCs on the intestinal microbiome alternation in aGVHD mice through 16S rRNA sequencing. Total of 1101 OTU were observed among the three groups (Figure 3A). Compared with Control group, aGVHD resulted in the reduction of  $\alpha$ -diversity of intestinal microbiome, manifested



**Figure 1.** HAMSCs prevented aGVHD in a humanized mouse model. (A) Schematic diagram of aGVHD induction and hAMSCs infusion. (B) Body weight change (n=10). (C) aGVHD clinical score (n=10). (D) Survival (n = 10). (E) H&E staining (magnification × 400) and histological score (n = 5). (F) Immunohistochemistry (magnification × 400) (n = 5). AGVHD target organs in the aGVHD group were collected when severe aGVHD occurred and target organs in the hAMSCs group were collected on day 6 after transplantation. \*\*  $P < .01$ , \*\*\*  $P < .001$ , \*\*\*\*  $P < .0001$ , based on the Student *t*-test.



**Figure 2.** HAMSCs improved intestinal barrier integrity in aGVHD mice. (A) The expression of ZO-1 protein in the intestine in situ. (B) The mRNA expression of ZO-1. (C) The expression of Occludin protein in the intestine in situ. (D) The mRNA expression of Occludin.  $n=5$  mice per group. Intestines in the aGVHD group were collected when severe aGVHD occurred and intestines in the hAMSCs group were collected on day 6 post-transplantation. \*\*  $P < .01$ , \*\*\*  $P < .001$ , \*\*\*\*  $P < .0001$ , ns, no significance, based on the one-way ANOVA and Tukey's post hoc test. ##  $P < .01$ , based on the Student  $t$ -test.



**Figure 3.** HAMSCs modulated intestinal microbiome in aGVHD mice. (A) Venn diagram. The  $\alpha$ -diversity of intestinal microbiome measured by OTUs (B) and Chao 1 index (C). The  $\beta$ -diversity of intestinal microbiome measured by PCoA based on Bray\_Curtis (D) and unweighted\_unifrac (E). (F) The  $\beta$ -diversity of intestinal microbiome measured by principal component analysis. Microbial community composition at the phylum (G), family (H), and genus (I) level. Relative bacterial abundance among groups at the phylum (J), family (K), and genus (L) level.  $N=5$  mice per group. Feces in the aGVHD group were collected when severe aGVHD occurred, and feces in the hAMSCs group and control group were collected on day 6 after transplantation. \*  $P < .05$ , \*\*  $P < .01$ , \*\*\*  $P < .001$ , ns, no significance, based on the one-way ANOVA and Tukey's post hoc test. #  $P < .05$ , based on the student's  $t$ -test. PCoA, Principal coordinate analysis; PC, Principal component.

by a decrease in OTU and Chao1 index, which were increased after treatment with hAMSCs (Figure 3B,C). We then analyzed the structure of intestinal microbiome based on Bray\_Curtis and unweighted\_unifrac. Principal coordinate analysis (PCoA) showed that the intestinal microbiome composition of the control, aGVHD, and hAMSCs groups had obvious cluster separation (Figure 3D,E). The results from principal component analysis were consistent with PCoA (Figure 3F). The alternation of species abundance at the phylum, family and genus levels was further observed to assess the differences in intestinal microbiome composition among groups. At the phylum level, aGVHD brought about a dramatic decrease in the relative abundance of *Firmicutes* and an increase in the relative abundance of *Bacteroidetes* (Figure 3G,J). Additionally, the ratio of *Firmicutes* to *Bacteroidetes* was dramatically decreased in the aGVHD group compared with the control group (Figure 3G, J). At the family and genus level, aGVHD also led to an increase in the relative abundance of *Prevotellaceae* and *Alloprevotella* and a decrease in the relative abundance of *Saccharimonadaceae* and *Candidatus\_Saccharimonas* (Figure 3H,I,K,L). HAMSCs treatment could regulate the relative abundance of the above intestinal microbiome caused by aGVHD to varying degrees (Figure 3G-L). Finally, we performed ANOVA-like differential expression-2 (ALDEx2) analyses to identify significantly different abundant microbiome among the control, aGVHD, and hAMSCs groups at the phylum, class, order, family, genus, and species levels (Tables 1,2; Supplementary Tables S1-S4). ALDEx2 analysis showed significant differences in the abundance of *Bacteroidetes*, *Patescibacteria*, *Tenericutes*, *Cyanobacteria*, and *Epsilonbacteraeota* among groups at the phylum level (Table 1). At the genus level, the genera that differed among groups were *Candidatus\_Arthromitus*, *Alloprevotella*, *Bacteroides*, *Odoribacter*, *Ruminococcaceae\_UCG-014*, *Rikenella*, *Clostridium\_sensu\_stricto\_1*, *Prevotellaceae\_UCG-001*, *Ruminococcus\_1*, *Ruminiclostridium\_9* and *Ruminiclostridium* (Table 2). ALDEx2 analysis at the phylum and genus level between control group and aGVHD group, as well as between aGVHD group and hAMSCs group were shown in Supplementary Figure S2 and Supplementary Tables S5 through S8. Additionally, analysis of compositions of microbiomes was also applied to discover species with significant differences among groups at different taxa (Supplementary

**Table 1**

ALDEx2 Analysis Among Groups at the Phylum Level

| Phylum                    | glm.ep   | glm.eBH  |
|---------------------------|----------|----------|
| <i>Firmicutes</i>         | 0.780296 | 0.794647 |
| <i>Bacteroidetes</i>      | 0.011811 | 0.023372 |
| <i>Proteobacteria</i>     | 0.209897 | 0.274771 |
| <i>Verrucomicrobia</i>    | 0.203919 | 0.268991 |
| <i>Patescibacteria</i>    | 4.30E-06 | 1.67E-05 |
| <i>Tenericutes</i>        | 7.87E-07 | 5.26E-06 |
| <i>Cyanobacteria</i>      | 0.000673 | 0.001688 |
| <i>Deferribacteres</i>    | 0.056357 | 0.093042 |
| <i>Actinobacteria</i>     | 0.713111 | 0.753541 |
| <i>Epsilonbacteraeota</i> | 8.20E-05 | 0.000238 |

glm.ep, p value using glm ANOVA test; glm.eBH, p value using glm ANOVA test and Benjamini-Hochberg correction.

Tables S9-S14). These findings indicated that hAMSCs could regulate the intestinal microecology, which was manifested by repairing the damage of intestinal barrier structure and function, as well as restoring the disorder of intestinal microbiome both caused by aGVHD.

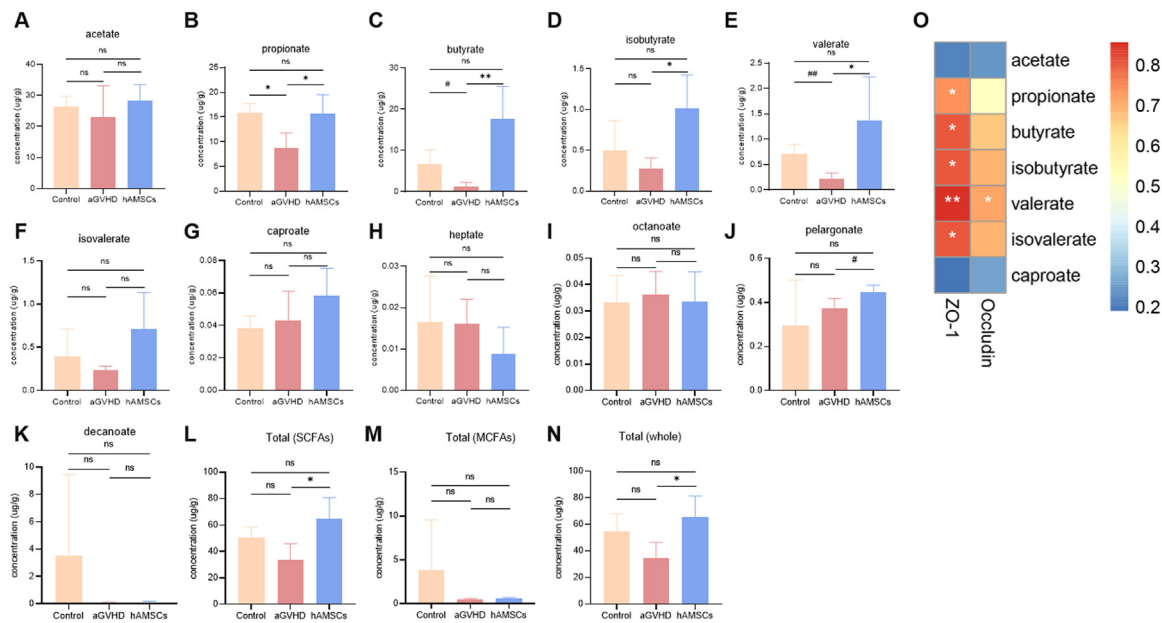
**Table 2**

ALDEx2 Analysis Among Groups at the Genus Level

| Genus                              | glm.ep   | glm.eBH     |
|------------------------------------|----------|-------------|
| <i>Lactobacillus</i>               | 0.104509 | 0.158301    |
| <i>Candidatus_Arthromitus</i>      | 9.80E-05 | 0.000665    |
| <i>Alloprevotella</i>              | 2.49E-07 | 6.43E-06    |
| <i>Bacteroides</i>                 | 0.002933 | 0.009675    |
| <i>Alistipes</i>                   | 0.000137 | 0.000831    |
| <i>Odoribacter</i>                 | 0.000376 | 0.001949    |
| <i>Ruminococcaceae_UCG-014</i>     | 0.005247 | 0.014701    |
| <i>Rikenella</i>                   | 6.33E-09 | 3.77E-07    |
| <i>Clostridium_sensu_stricto_1</i> | 0.001428 | 0.006038    |
| <i>Candidatus_Saccharimonas</i>    | 6.79E-05 | 0.000488    |
| <i>Prevotellaceae_UCG-001</i>      | 0.002724 | 0.009116    |
| <i>Ruminococcus_1</i>              | 6.11E-06 | 7.09E-05    |
| <i>Parabacteroides</i>             | 0.002068 | 0.007616    |
| <i>Ruminiclostridium_9</i>         | 0.002797 | 0.009255    |
| <i>Roseburia</i>                   | 3.08E-06 | 4.04E-05    |
| <i>Lachnoclostridium</i>           | 0.017888 | 0.040817    |
| <i>Ruminiclostridium_6</i>         | 0.074073 | 0.122051    |
| <i>Ruminococcaceae_UCG-013</i>     | 0.00013  | 0.000803    |
| <i>Ruminiclostridium</i>           | 0.011535 | 0.027361    |
| <i>Tyzzereella_3</i>               | 2.87E-05 | 0.000240642 |
| <i>Helicobacter</i>                | 0.000116 | 0.00043     |
| <i>Enterococcus</i>                | 0.086722 | 0.13571     |
| <i>ASF356</i>                      | 6.69E-05 | 0.000369421 |
| <i>Staphylococcus</i>              | 0.168017 | 0.222758    |
| <i>Streptococcus</i>               | 0.160554 | 0.216669    |

glm.ep, p value using glm ANOVA test; glm.eBH, p value using glm ANOVA test and Benjamini-Hochberg correction





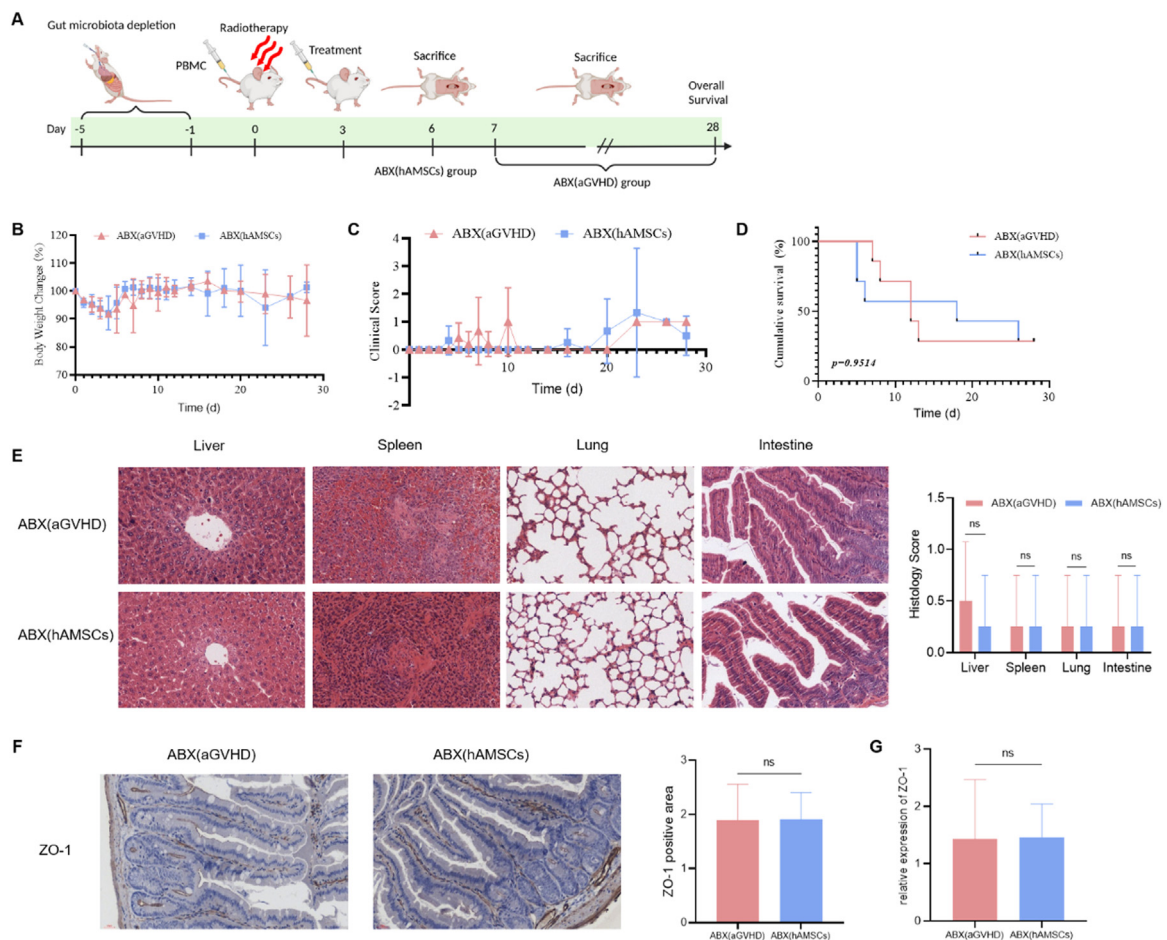
**Figure 4.** HAMSCs increased metabolite SCFAs to repair intestinal barrier disruption in aGVHD mice. (A–N) SCFAs and MCFAs concentration. (O) Correlation analysis between SCFAs and tight junction proteins. N = 3 mice in the Control group, and n = 4 mice in the aGVHD and hAMSCs groups. Colon contents in the aGVHD group were collected when severe aGVHD occurred and colon contents in the hAMSCs group were collected on day 6 post-transplantation. \*  $P < .05$ , \*\*  $P < .01$ , ns, no significance, based on the one-way ANOVA and Tukey's post hoc test. #  $P < .05$ , ##  $P < .01$ , based on the Student  $t$ -test. SCFAs, short-chain fatty acids; MCFAs, medium-chain fatty acids. Whole represents the total amount of both SCFAs and MCFAs.

### Intestinal Microbiome Gene Functions Were Associated With HAMSCs Infusion in aGVHD Mice

PICRUSt II function prediction analysis based on the Kyoto Encyclopedia of Genes and Genomes was carried out to infer the microbial gene functions. As shown in [Supplementary Figure S2](#), the intestinal microbiome of aGVHD influenced the pathway of “biosynthesis of vancomycin group antibiotics,” “D-glutamine and D-glutamate metabolism,” “secondary bile acid biosynthesis,” “D-alanine metabolism,” “peptidoglycan biosynthesis,” “mismatch repair,” “aminoacyl-tRNA biosynthesis,” “pentose phosphate pathway,” “ribosome,” “homologous recombination,” “lysine biosynthesis,” “fatty acid biosynthesis,” “lysine biosynthesis,” and “cell cycle-Caulobacter”. HAMSCs therapy could regulate the disorder of pathway changes caused by aGVHD, including “D-glutamine and D-glutamate metabolism,” “D-alanine metabolism,” “secondary bile acid biosynthesis,” “peptidoglycan biosynthesis,” “mismatch repair,” “aminoacyl-tRNA biosynthesis,” “pentose phosphate pathway,” “ribosome,” “fatty acid biosynthesis,” and “lysine biosynthesis.” The above results indicate that intestinal microbiome might participate in the preventive effect of hAMSCs on aGVHD.

### HAMSCs Increased Metabolites SCFAs To Repair Intestinal Barrier Disruption in aGVHD Mice

We detected and analyzed the concentration of MCFAs and SCFAs in the colon contents through targeted metabolomics based on GC-MS. As shown in [Figure 4A](#) through [N](#), the fatty acid profiles of mice in the control, aGVHD, and hAMSCs groups are nearly completely different. In terms of SCFAs, the occurrence of aGVHD resulted in the reduction of propionate, butyrate, and valerate. There was no significant difference in the concentration of MCFAs, including heptate, octanoate, pelargonate, and decanoate between the control and aGVHD groups. HAMSCs treatment dramatically increased the amounts of propionate, butyrate, isobutyrate, and valerate. Although the amounts of acetate, isovalerate, and caproate between the aGVHD and hAMSCs groups were not significantly different, total SCFA concentration in hAMSCs-treated mice was higher than that in aGVHD mice. Although the hAMSCs group harbored higher concentration of pelargonate when compared to the aGVHD group, there was no significant difference in the amounts of heptate, octanoate, and decanoate, as well as total MCFAs. Of note, after hAMSCs treatment, the butyrate concentration increased the most significantly, increasing by more than 10 times. Because SCFAs



**Figure 5.** The preventive effect of hAMSCs on aGVHD disappeared after gut microbiota depletion. (A) Schematic diagram of gut microbiota depletion and aGVHD induction. (B) Body weight change (n = 7). (C) aGVHD clinical score (n = 7). (D) Survival (n = 7). (E) H&E staining (magnification  $\times 400$ ) and histological score (n = 4). (F) The expression of ZO-1 protein in the intestine in situ (n = 4). (G) The mRNA expression of ZO-1 (n = 4). aGVHD target organs in the ABX(aGVHD) group and ABX(hAMSCs) group were collected on day 6 after PBMC injection. ns, no significance, based on the Student *t*-test. PBMC, peripheral blood mononuclear cells.

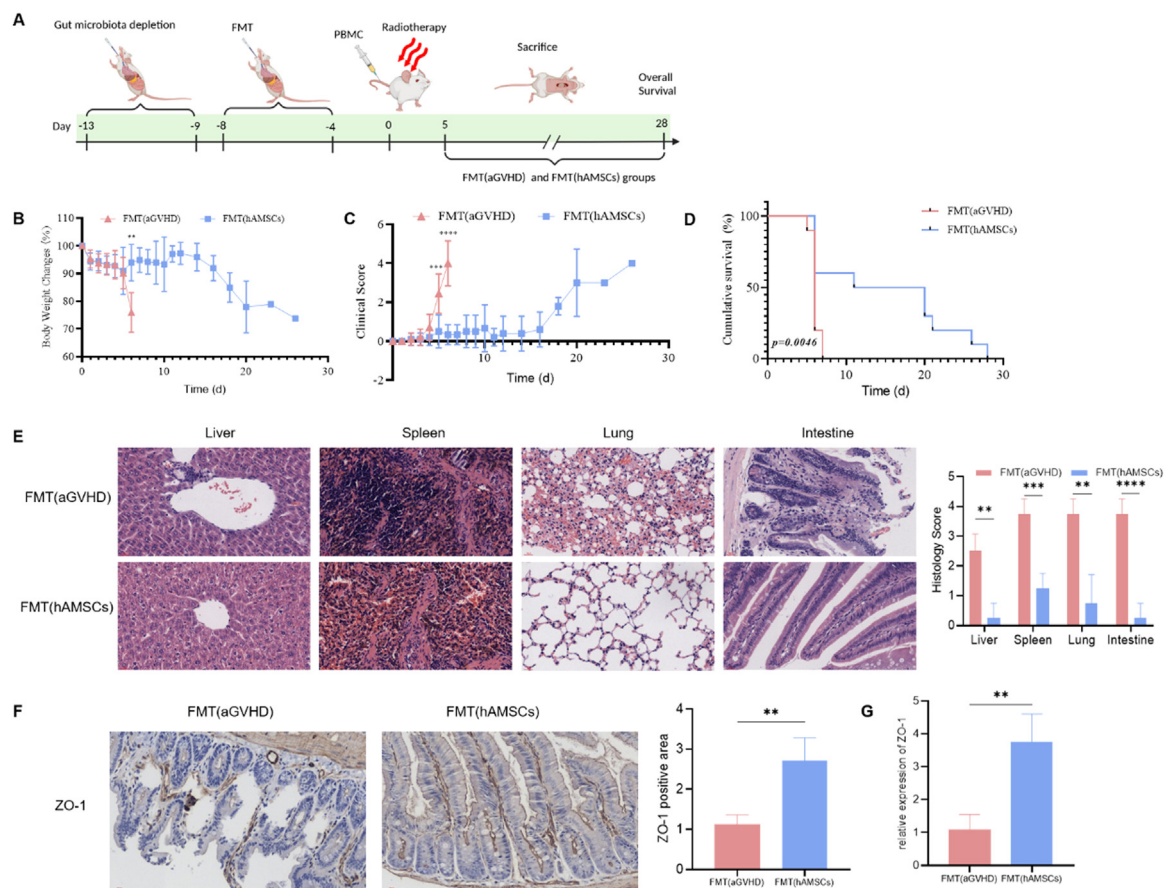
perform a significant role in maintaining the integrity of intestinal barrier and the changes of SCFAs are most obvious after hAMSCs treatment, we analyzed the correlation between the concentration of SCFAs and the mRNA level of TJs to find the latent association between SCFAs and intestinal barrier function. As shown in Figure 4O, the amounts of propionate, butyrate, isobutyrate, valerate, and isovalerate were positively correlative with the mRNA level of ZO-1. Meanwhile, there was a positive correlation between valerate concentration and the occludin mRNA level. Collectively, hAMSCs administration increased intestinal microbiome-derived metabolites SCFAs, which might be associated with intestinal barrier function.

### **HAMSCs Prevented aGVHD in an Intestinal Microbiome-Dependent Manner**

To explore whether intestinal microbiome got involved in the preventive effect of hAMSCs on

aGVHD, quadruple antibiotic cocktails were conducted intragastrically on NPG mice for gut microbiota depletion before radiotherapy (Figure 5A represented the schematic diagram). After gut microbiota depletion, body weight loss (Figure 5B), aGVHD clinical score (Figure 5C), survival (Figure 5D), or histology score of target organs (Figure 5E) in mice between ABX(aGVHD) and the ABX(hAMSCs) groups displayed no distinguishable difference. Furthermore, the protein and mRNA expression of ZO-1 were also not significantly different between the 2 groups (Figure 5F,G). We then performed FMT experiment to further verify whether the inhibition of aGVHD after hAMSCs administration relied on the intestinal microbiome (Figure 6A represented the schematic diagram). The results of FMT experiment showed that mice in the FMT(hAMSCs) group harbored a significant reduction in aGVHD symptoms, reflected in reduced weight loss





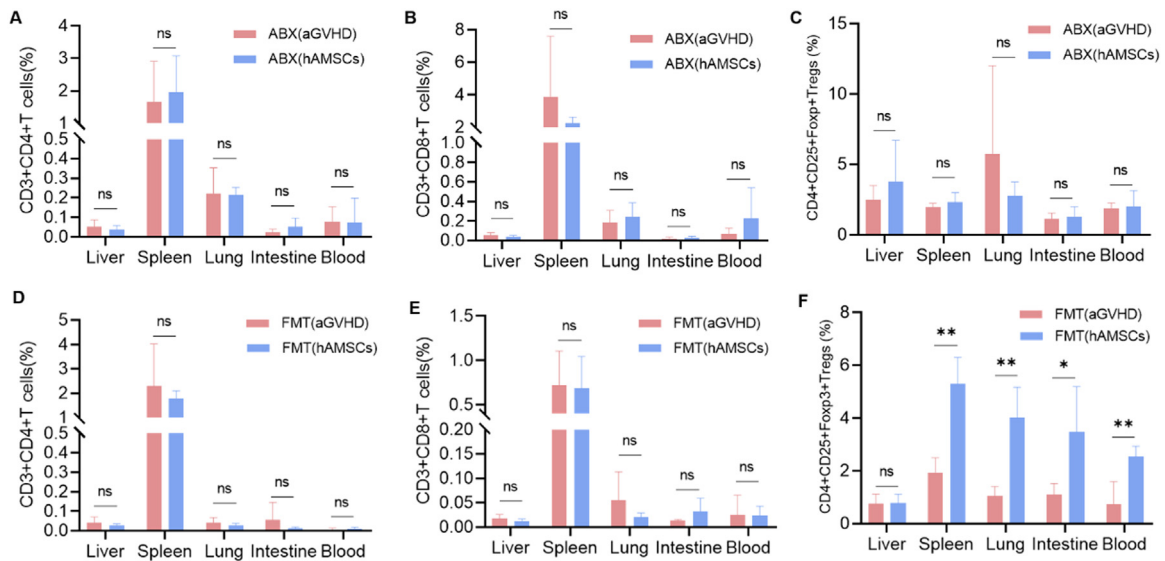
**Figure 6.** Fecal microbial transplantation ameliorated aGVHD. (A) Schematic diagram of fecal microbial transplantation and aGVHD induction. (B) Body weight change ( $n = 10$ ). (C) aGVHD clinical score ( $n = 10$ ). (D) Survival ( $n = 10$ ). (E) H&E staining (magnification  $\times 400$ ) and histological score ( $n = 4$ ). (A) The expression of ZO-1 protein in the intestine in situ ( $n = 4$ ). (B) The mRNA expression of ZO-1 ( $n = 4$ ). aGVHD target organs in the FMT(aGVHD) group and FMT(hAMSCs) group were collected on day 6 after PBMC injection. \*\*  $P < .01$ , \*\*\*  $P < .001$ , \*\*\*\*  $P < .0001$ , based on the Student  $t$ -test. FMT, fecal microbiota transplantation; PBMC, peripheral blood mononuclear cells.

(Figure 6B), decreased clinical aGVHD score (Figure 6C), and significantly prolonged survival (Figure 6D). The results from H&E staining showed slighter tissue damage and less lymphocyte infiltration in the FMT(hAMSCs) group compared to FMT(aGVHD) group (Figure 6E). Additionally, transplantation with feces from hAMSCs-treated mice remarkably promoted the expression of ZO-1 protein and mRNA (Figure 6F,G). These results showed that intestinal microbiome in mice treated with hAMSCs played an important role in mitigating aGVHD. Collectively, the above findings indicated that the preventive effect of hAMSCs on aGVHD was dependent on intestinal microbiome.

#### **HAMSCs Regulated the Balance of Tregs and Effector T Cells in a Partially Intestinal Microbiome-Dependent Manner**

Our previous studies showed that hAMSCs could regulate the balance of Tregs and effector T

cells (Teffs) to mitigate inflammatory response in aGVHD mice [11]. We further investigated whether the improvement of T-cell immune balance in aGVHD mice after hAMSCs infusion was dependent on the intestinal microbiome. Interestingly, there was no significant difference in the frequencies of CD3+CD4+T cells, CD3+CD8+T cells and Tregs between ABX(aGVHD) and ABX (hAMSCs) groups (Figure 7A–C and Supplementary Figures S3–S5). After FMT, the FMT(aGVHD) and FMT(hAMSCs) groups displayed indistinguishable frequencies of CD3+CD4+T cells and CD3+CD8+T cells (Figure 7D,E; Supplementary Figures S6–S8). However, the frequencies of Tregs in FMT(hAMSCs) group were higher than those in FMT(aGVHD) group in peripheral blood and target organs apart from the liver (Figure 7F; Supplementary Figure S8). In general, hAMSCs regulated Tregs/Teffs balance in a partially intestinal microbiome-dependent manner.



**Figure 7.** The frequencies of T cells in mice after gut microbiota depletion and FMT. CD3+CD4+T cells (A), CD3+CD8+T cells (B), and Tregs (C) from ABX(aGVHD) and ABX(hAMSCs) groups. CD3+CD4+T cells (D), CD3+CD8+T cells (E), and Tregs (F) from FMT(aGVHD) and FMT(hAMSCs) groups. N = 4 mice per group. AGVHD target organs and peripheral blood in all groups were collected on day 6 after PBMC injection. \*  $P < .05$ , \*\*  $P < .01$ , ns, no significance, based on the Student *t*-test.

## DISCUSSION

In the present study, we used a humanized aGVHD mouse model to explore the preventive effect of hAMSCs on aGVHD and the benefits of hAMSCs on the intestinal microbiome during this process. First, we demonstrated that hAMSCs administration attenuated aGVHD in vivo, as evidenced by less weight loss, alleviated aGVHD symptoms, and longer survival. Then we presented the fact that hAMSCs treatment reshaped intestinal homeostasis including intestinal barrier dysfunction and intestinal microbiome dysbiosis induced by aGVHD. Furthermore, hAMSCs increased the concentration of intestinal microbiome-derived metabolites SCFAs, especially butyrate. Moreover, we also demonstrated that hAMSCs regulated Tregs/Teffs balance to prevent aGVHD in a partially intestinal microbiome-dependent manner through gut microbiota depletion and fecal microbial transplantation experiments.

Acute GVHD, as a common seriously lethal complication after allo-HSCT, is a complex and abnormal immune response to the recipient's multiple tissues and organs caused by the mismatch of human leukocyte antigen between donor and recipient to trigger T lymphocyte activation and cytokine cascade [17,18]. Therefore immunosuppressive agents are the first choice for aGVHD treatment, and some new medicaments such as Janus kinase (JAK) 1/2 inhibitor ruxolitinib,

TNF- $\alpha$  inhibitors, and IL-2 receptor antibodies have been put into clinical use as a second-line treatment, but a proportion of patients, especially those with intestinal aGVHD, still suffer treatment resistance [19,20]. In fact, the imbalance of intestinal homeostasis could not be corrected completely after immunosuppressive treatment. Intestinal homeostasis refers to a dynamic equilibrium state, which is mainly composed of the host (intestinal mucosal and immunological barrier), intestinal internal environment (including intestinal microbiome), nutrition, and metabolites (including SCFAs, BAs, and indoles) derived from intestinal microbiome. During allo-HSCT, intestinal homeostasis including the intestinal barrier and intestinal microbiome could be disrupted by the conditioning regimen and antibiotics, leading to delayed wound healing, enhanced immune response, and continuous tissue damage, whereas immunosuppressive therapy may not be able to fully recover them [5]. Recently, researchers pointed out that remodeling intestinal homeostasis may be a new direction of aGVHD integrated management [5].

In addition to the reported anti-inflammatory and tissue repair capabilities of MSC [21,22], recent studies have found that MSC appeared to normalize the intestinal microbiome in vivo. Soontarak et al. [23] demonstrated that MSC administration stimulated intestinal epithelial cell proliferation, restored intestinal microbiome

composition, and ultimately reshaped intestinal homeostasis in colitis mouse models. Li et al. [24] found that MSC could regulate the interaction between the intestinal microbiome and host immunity to alleviate rheumatoid arthritis in rats. In a sepsis model, MSC could also reverse intestinal microbiome disorder to improve acute lung injury caused by sepsis [25]. In the present study, significantly alleviated aGVHD was achieved with hAMSCs infusion, which is consistent with the previous studies [11–13]. As expected, we also observed that hAMSCs administration had the ability to mitigate the damage of intestinal barrier structure caused by aGVHD, and repair intestinal barrier integrity by increasing TJs. The 16S rRNA sequencing was performed to investigate the intestinal microbiome alterations after hAMSCs administration. We observed that the occurrence of aGVHD dramatically reduced the diversity and altered the composition of intestinal microbiome in mice. During allo-HSCT, patients receive comprehensive treatment such as radiotherapy, chemotherapy, and antibiotics, which results in the disorder of intestinal microbiome composition and the loss of intestinal microbiome diversity, and meanwhile, the reduction of intestinal microbiome diversity and the expansion of pathogenic bacteria have been proven to be positively connected with the risk of aGVHD [6]. Taur et al. [26] and Jenq et al. [27] observed a reduction in the diversity of intestinal microbiome in both the aGVHD mouse and patients with aGVHD, indicating that intestinal microbiome plays an essential role in the occurrence and development of aGVHD. Similar to the previous study reported by Eriguchi et al. [28], our results showed that *Firmicutes* and *Bacteroidetes* were the 2 most phyla before aGVHD, whereas the ratio of *Firmicutes* to *Bacteroidetes* was reduced after aGVHD. After hAMSCs treatment, the diversity of intestinal microbiome in aGVHD mice was increased obviously, and the changes in intestinal microbiome composition caused by aGVHD were significantly regulated, with genera *Ruminococcus\_1* and *Clostridium\_sensu\_stricto\_1* increased obviously. *Ruminococcus* and *Clostridium* have been proven to effectively enhance intestinal barrier, produce SCFAs and reduce inflammation [29,30]. Collectively, hAMSCs administration could restore intestinal microbiome dysbiosis caused by aGVHD and normalize it to be beneficial to the body.

Studies have shown that intestinal microbiome has a profound impact on host immune response through bacteria composition and its metabolites [31]. Recently, the relationship between intestinal

microbiome-derived metabolites and pathophysiological mechanisms of diseases has attracted more attention. In the maintenance of intestinal homeostasis, SCFAs, BAs and tryptophan metabolites play crucial roles among various metabolites [32,33]. SCFAs is an important metabolite derived from intestinal microbiome to play the role in immune regulation. Butyrate, propionate, and acetate are the main energy sources of intestinal epithelial cells, which can retain intestinal barrier integrity and inhibit intestinal inflammation, thus affecting the host immune response [34]. Similar to SCFAs, MCFAs can activate peroxisome proliferator-activated receptor- $\gamma$  to display anti-inflammatory activity [35]. In a mouse model, Mathewson et al. [36] found that the concentration of butyrate in the intestinal tissue decreased, and supplemented butyrate can protect intestinal epithelial cells from allo-HSCT induced immune attack mediated by alloreactive T cells, thereby reducing aGVHD severity, which resulted from the high production of SCFAs by *Clostridium* strains. In the present study, we showed that hAMSCs treatment increased SCFAs especially butyrate concentration in colon contents, and meanwhile, *Clostridium\_sensu\_stricto\_1* displayed a relative enrichment in the hAMSCs-treated mice. Spearman correlation analysis revealed that SCFAs concentrations had positive connections with the mRNA expression level of tight junction proteins. Thus we speculated that hAMSCs may repair intestinal barrier damage by remodeling the intestinal microbiome and increasing the relative abundance of SCFAs producing bacteria, thus preventing aGVHD, which needs to be further confirmed.

Since intestinal microbiome dysbiosis participants in the pathogenesis of aGVHD, we have found that intestinal microbiome could be altered after hAMSCs infusion, so we will next explore the role of intestinal microbiomes in the preventive process of hAMSCs alleviating aGVHD. Interestingly, after gut microbiota depletion, the anti-aGVHD effect of hAMSCs seemed to disappear, demonstrating that the intestinal microbiome got involved in the preventive process of hAMSCs alleviating aGVHD. Furthermore, the FMT experiment showed that the preventive effect of hAMSCs on aGVHD appeared after FMT, indicating that intestinal microbiome was involved in the pathophysiological process of hAMSCs ameliorating aGVHD. These results demonstrated that hAMSCs prevented aGVHD in an intestinal microbiome-dependent manner.

During allo-HSCT, the conditioning regimen and antibiotic treatment lead to the intestinal



microbiome dysbiosis, which could affect the immune balance between T effector cells and Tregs and thus cause excessive inflammatory response in the intestine [37]. Our previous study has shown that hAMSCs inhibit aGVHD by improving the Tregs/ Teffs balance in the aGVHD target organs and peripheral blood [11,14]. Subsequently, we explored whether the improvement of T cell immune imbalance after hAMSCs administration was dependent on intestinal microbiome. Strikingly, after gut microbiota depletion, there was no significant difference in the frequencies of CD3+CD4+T cells, CD3+CD8+T cells, and Tregs between the ABX(aGVHD) and ABX (hAMSCs) groups. Additionally, the CD3+CD4+T and CD3+CD8+T cells displayed no significant difference between the FMT(aGVHD) and FMT (hAMSCs) groups, but the frequency of Tregs was dramatically higher in the FMT(hAMSCs) group than those in FMT(aGVHD) group in the spleen, lung, intestine, and peripheral blood. Our findings indicate that hAMSCs regulate the Tregs/Teffs balance in a partially intestinal microbiome-dependent manner. To the best of our knowledge, this is the first study to demonstrate that hAMSCs prevent aGVHD in an intestinal microbiome-dependent manner in an aGVHD mouse model and to clarify the potential mechanism between hAMSCs regulating T-cell response and intestinal microbiome.

There remain several limitations for improvement in the following experiment. First, the n number in the experiments is relatively small. Second, because of the limitation of fecal volume of aGVHD mice, 16S rRNA-sequencing and GC-MS-targeted metabolomics detection samples are different samples from the same group, which makes it impossible to carry out correlation analysis on intestinal microbiome and its metabolite SCFAs to discover the potential mechanism between the them. Third, the role of SCFAs as important players participating in hAMSCs alleviation of aGVHD needs further investigation. Finally, it is necessary to further explore the underlying mechanisms by which hAMSCs regulate intestinal microbiome.

In conclusion, our study indicated that hAMSCs prevented aGVHD in an intestinal microbiome-dependent manner. The potential preventive mechanisms were related to the repair of the intestinal barrier, restoring intestinal microbiome and increasing intestinal microbiome metabolites SCFAs, especially butyrate concentration, to reshape intestinal homeostasis. It might shed light on a new mechanism of hAMSCs inhibiting

aGVHD and promote the development of new prophylaxis regimens for aGVHD prevention.

## ACKNOWLEDGMENTS

*Financial disclosure:* Supported by the National Natural Science Foundation of China (81870145, 82200234), Science and Technology Planning Project of Guangzhou (201904010481) and President Foundation of Nanfang Hospital, Southern Medical University (2022B008). Figure 1A, Figure 5A and Figure 6A were created with BioRender.com.

*Conflict of interest statement:* There are no conflicts of interest to report.

*Authorship statement:* B.H.P., and Y.G. designed the experiments. X.Y.B. performed the experiments, analyzed data and wrote the paper. W.F.P, L. P.L., J.H.W. and Z.Y. provided critical revisions for the manuscript. All authors reviewed and approved the final manuscript. All authors have made substantial contributions to the conception or design of this work and the acquisition, analysis, and interpretation of data for the work. All authors drafted the manuscript or revised it critically for important intellectual content and provided final approval of the version to be published. All authors confirm that they had full access to all the data in the study and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

## SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.jtct.2023.11.005.

## REFERENCES

1. Martinez-Cibrian N, Zeiser R, Perez-Simon JA. Graft-versus-host disease prophylaxis: pathophysiology-based review on current approaches and future directions. *Blood Rev.* 2021;48: 100792.
2. Penack O, Marchetti M, Ruutu T, et al. Prophylaxis and management of graft versus host disease after stem-cell transplantation for haematological malignancies: updated consensus recommendations of the European Society for Blood and Marrow Transplantation. *Lancet Haematol.* 2020;7:e157–e167.
3. Morgan XC, Huttenhower C. Meta'omic analytic techniques for studying the intestinal microbiome. *Gastroenterology.* 2014;146:1437–1448.
4. Wang W, Xu S, Ren Z, Tao L, Jiang J, Zheng S. Application of metagenomics in the human gut microbiome. *World J Gastroenterol.* 2015;21:803–814.
5. Gao F, Wu H, Wang L, Zhao Y, Huang H. Altered intestinal microbiome and epithelial damage aggravate intestinal graft-versus-host disease. *Gut Microbes.* 2023;15: 2221821.
6. Hong T, Wang R, Wang X, et al. Interplay between the intestinal microbiota and acute graft-versus-host

- disease: experimental evidence and clinical significance. *Front Immunol.* 2021;12: 644982.
7. Blazar BR, Murphy WJ, Abedi M. Advances in graft-versus-host disease biology and therapy. *Nat Rev Immunol.* 2012;12:443–458.
  8. Markov A, Thangavelu L, Aravindhan S, et al. Mesenchymal stem/stromal cells as a valuable source for the treatment of immune-mediated disorders. *Stem Cell Res Ther.* 2021;12:192.
  9. Liu Q, Huang Q, Wu H, et al. Characteristics and therapeutic potential of human amnion-derived stem cells. *Int J Mol Sci.* 2021;22:970.
  10. Umezawa A, Hasegawa A, Inoue M, et al. Amnion-derived cells as a reliable resource for next-generation regenerative medicine. *Placenta.* 2019;84:50–56.
  11. Gao Y, Li W, Bu X, et al. Human amniotic mesenchymal stem cells inhibit aGVHD by regulating balance of Treg and T effector cells. *J Inflamm Res.* 2021;14:3985–3999.
  12. Yamahara K, Harada K, Ohshima M, et al. Comparison of angiogenic, cytoprotective, and immunosuppressive properties of human amnion- and chorion-derived mesenchymal stem cells. *PLOS One.* 2014;9:e88319.
  13. Tago Y, Kobayashi C, Ogura M, et al. Human amnion-derived mesenchymal stem cells attenuate xenogeneic graft-versus-host disease by preventing T cell activation and proliferation. *Sci Rep-UK.* 2021;11:2406.
  14. Bu X, Wang J, Yin Z, et al. Human amniotic mesenchymal stem cells alleviate aGVHD after allo-HSCT by regulating interactions between gut microbiota and intestinal immunity. *Stem Cell Rev Rep.* 2023;19:1370–1383.
  15. Cooke KR, Kobzik L, Martin TR, et al. An experimental model of idiopathic pneumonia syndrome after bone marrow transplantation: I. The roles of minor H antigens and endotoxin. *Blood.* 1996;88:3230–3239.
  16. Yañez R, Lamana ML, García-Castro J, Colmenero I, Ramírez M, Bueren JA. Adipose tissue-derived mesenchymal stem cells have in vivo immunosuppressive properties applicable for the control of the graft-versus-host disease. *Stem Cells.* 2006;24:2582–2591.
  17. Ball LM, Egeler RM. Acute GvHD: pathogenesis and classification. *Bone Marrow Transplant.* 2008;41(Suppl 2): S58–S64.
  18. Ferrara JL, Levine JE, Reddy P, Holler E. Graft-versus-host disease. *Lancet.* 2009;373:1550–1561.
  19. Malard F, Huang XJ, Sim J. Treatment and unmet needs in steroid-refractory acute graft-versus-host disease. *Leukemia.* 2020;34:1229–1240.
  20. Luo C, Huang X, Wei L, et al. Second-line therapy for patients with steroid-refractory aGVHD: systematic review and meta-analysis of randomized controlled trials. *Front Immunol.* 2023;14:1211171.
  21. Singer NG, Caplan AI. Mesenchymal stem cells: mechanisms of inflammation. *Ann Rev Pathol.* 2011;6:457–478.
  22. Wang Y, Chen X, Cao W, Shi Y. Plasticity of mesenchymal stem cells in immunomodulation: pathological and therapeutic implications. *Nat Immunol.* 2014;15:1009–1016.
  23. Soontarak S, Chow L, Johnson V, et al. Mesenchymal stem cells (MSC) derived from induced pluripotent stem cells (iPSC) equivalent to adipose-derived MSC in promoting intestinal healing and microbiome normalization in mouse inflammatory bowel disease model. *Stem Cell Transl Med.* 2018;7:456–467.
  24. Li X, Lu C, Fan D, et al. Human umbilical mesenchymal stem cells display therapeutic potential in rheumatoid arthritis by regulating interactions between immunity and gut microbiota via the aryl hydrocarbon receptor. *Front Cell Dev Biology.* 2020;8:131.
  25. Sun J, Ding X, Liu S, Duan X, Liang H, Sun T. Adipose-derived mesenchymal stem cells attenuate acute lung injury and improve the gut microbiota in septic rats. *Stem Cell Res Ther.* 2020;11:384.
  26. Taur Y, Jenq RR, Perales M, et al. The effects of intestinal tract bacterial diversity on mortality following allogeneic hematopoietic stem cell transplantation. *Blood.* 2014;124:1174–1182.
  27. Jenq RR, Ubeda C, Taur Y, et al. Regulation of intestinal inflammation by microbiota following allogeneic bone marrow transplantation. *J Exp Med.* 2012;209:903–911.
  28. Eriguchi Y, Takashima S, Oka H, et al. Graft-versus-host disease disrupts intestinal microbial ecology by inhibiting Paneth cell production of  $\alpha$ -defensins. *Blood.* 2012;120:223–231.
  29. Maruyama S, Matsuoka T, Hosomi K, et al. Classification of the occurrence of dyslipidemia based on gut bacteria related to barley intake. *Front Nutr.* 2022;9: 812469.
  30. Guo P, Zhang K, Ma X, He P. Clostridium species as probiotics: potentials and challenges. *J Anim Sci Biotechnol.* 2020;11:24.
  31. Han H, Yi B, Zhong R, et al. From gut microbiota to host appetite: gut microbiota-derived metabolites as key regulators. *Microbiome.* 2021;9:162.
  32. Wang J, Zhu N, Su X, Gao Y, Yang R. Gut-microbiota-derived metabolites maintain gut and systemic immune homeostasis. *Cells.* 2023;12(5):93.
  33. Nicolas GR, Chang PV. Deciphering the chemical lexicon of host-gut microbiota interactions. *Trends Pharmacol Sci.* 2019;40:430–445.
  34. Masetti R, Zama D, Leardini D, et al. Microbiome-derived metabolites in allogeneic hematopoietic stem cell transplantation. *Int J Mol Sci.* 2021;22:1197.
  35. Lavelle A, Sokol H. Gut microbiota-derived metabolites as key actors in inflammatory bowel disease. *Nat Rev Gastroenterol Hepatol.* 2020;17:223–237.
  36. Mathewson ND, Jenq R, Mathew AV, et al. Gut microbiome-derived metabolites modulate intestinal epithelial cell damage and mitigate graft-versus-host disease. *Nat Immunol.* 2016;17:505–513.
  37. Lin D, Hu B, Li P, Zhao Y, Xu Y, Wu D. Roles of the intestinal microbiota and microbial metabolites in acute GVHD. *Exp Hematol Oncol.* 2021;10:49.